Sphinx: Certificateless Conditional Privacy-Preserving Authentication With Secure Transmission for VANETs

Mengjie Zhou, Chao Lin, *Member, IEEE*, Shengmin Xu, *Member, IEEE*, and Wei Wu

***Abstract*—Conditional privacy-preserving authentication (CPPA) has been widely studied to ensure authentication, anonymity, and traceability in vehicle ad hoc networks (VANETs). Among CPPA schemes, certificateless CPPA (CL-CPPA) offers natural advantages for VANETs by avoiding the complex certificate management of PKI-based solutions and the key escrow problem of identity-based systems. However, many existing CL-CPPA schemes overlook secure message communication. When confidentiality is required, they often rely on additional encryption, which adds communication overhead comparable to that of certificate transmission in PKI-based solutions. Anamorphic signatures (proposed at CRYPTO'23) can be adopted to address this issue by embedding sensitive messages within signatures, which eliminates the need for separate encryption. However, existing anamorphic signature schemes are symmetric, requiring extensive key agreements between two entities, which is impractical for the dynamic nature of VANETs. In this paper, we introduce a new cryptographic primitive, asymmetric anamorphic signatures (AAS), which enables secure message sharing in a public-key setting, thus eliminating the need for complex key agreements. We also provide a concrete construction of AAS based on a variant of the Hohenberger-Waters signature scheme. By integrating AAS with CL-CPPA, we propose *Sphinx*, a scheme that ensures authentication, anonymity, traceability, and confidentiality without the overhead of additional secure transmissions or complex key agreements. Our security proofs and performance evaluations demonstrate the practicality and efficiency of *Sphinx*, particularly its advantage in reducing communication overhead.**

***Index Terms*—VANETs, CPPA, certificateless, anamorphic signature.**

I. Introduction

CURRENTLY, the construction of smart cities is urgent, with road safety issues being particularly prominent. In Europe, the number of deaths due to road traffic accidents is as high as about 500 per week, a grim reality that prompted the European Union to release a road safety policy aimed at achieving a near-zero-fatality vision by 2050, further emphasizing the critical importance of road safety [1]. Against this, intelligent transportation systems (ITS) have emerged as the solution, significantly improving traffic safety and efficiency with advanced technologies such as vehicle ad hoc networks (VANETs) and autonomous driving [2]. Promoting the development of ITS, especially relying on VANETs, has become a key strategy to meet the challenges of building smart cities [3], [4]. Through this system, vehicles can broadcast basic safety messages (BSMs) or event-driven safety messages (ESMs) to neighboring nodes, enhancing vehicle awareness [5] and dynamically exchanging real-time traffic data.

While this approach improves traffic conditions and enhances road safety, the communication process may expose the identity information of the communicator. This limitation presents challenges for VANETs that require openness. Since secure identity authentication is a prerequisite for system security, identities need to be verified prior to all communications [6], [7], [8], [9]. Conditional privacy-preserving authentication (CPPA), as the core technology for realizing identity authentication between entities, is responsible for performing identity verification, privacy protection, and behavior constraints. It is a prerequisite for smooth communication and is used to protect the anonymity of honest users and to track the identity of malicious users. Over the past decades, many researchers have proposed various CPPA schemes, which can be broadly categorized into PKI-based, identity-based, and certificateless (CL) [10]. PKI-based solutions with complex certificate management and identity-based systems with key escrow do not apply to VANETs [11]. Specifically, key escrow means that the keys of vehicles are generated by KGC, which retains knowledge of these keys. Consequently, if the KGC is compromised, it can impersonate any vehicle and disseminate falsified traffic information, potentially resulting in severe disruption to the system. In this regard, the inherent properties of CL-CPPA confer natural advantages, as the private key of each vehicle is jointly generated by both the KGC and the vehicle itself, thereby mitigating the risks of key escrow [12].

However, existing CL-CPPA protocols mainly focus on anonymous authentication and identity-tracking functions without considering the secure transmission of messages (which is guaranteed by default through the additional encryption after signatures). Although the signcryption scheme [13] can simultaneously ensure the message's unforgeability and confidentiality and has a low computational cost, it needs to transmit the ciphertext in addition to the signature during

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the communication process. In this process, the additional communication overhead incurred by transmitting the ciphertext leads to system delay, which is detrimental to VANETs.

For detailed analysis, based on the dedicated short-range communication (DSRC) protocol used in VANETs, each vehicle propagates data in the range of 100–300 ms, and the rate of data transmission is usually maintained between 6–27 Mbps. This means that the amount of data transmitted is approximately between 75,000–337,500 Bytes per transmission [14]. However, existing schemes transmit not only public parameter and message signature pairs but also additional ciphertexts. Typically, for simple identity-based encryption schemes, the length of the ciphertext is often 2-4 times[1] the length of the plaintext [15], [16]. For attribute-based encryption (ABE) schemes, the length of the ciphertext is even longer, with its complexity closely related to the number of attributes or the complexity of the access policy [17].

As a result, existing solutions could not fully meet the stringent processing requirements of VANETs in terms of security authentication and confidential communications.

We are considering secretly embedding sensitive information in the signature transmission process to securely deliver messages while achieving authentication. The concept of "Anamorphic Signature" proposed on CRYPTO'23 [18], serves a similar purpose, whereby anamorphic messages are coded in the signature, which is no different from a normal signature and is more appropriate for the DSRC protocol in VANETs. It is worth noting that the single anamorphic signature transmission channel is limited, typically 32 Bytes. If the complete information is embedded into the anamorphic signature, multiple authentication messages are required, which will lead to inefficiency. Therefore, we only transmit sensitive data in an anamorphic channel and leave other information unclassified. Specifically, the data we send usually contains speed, location, timestamp, pseudonym, and signature. We distill sensitive data such as speed and location as anamorphic information, while timestamps, pseudonyms, and signatures are publicly available as metadata [19], [20], [21].

The introduction of the anamorphic signature in CPPA schemes can effectively address the additional secure transmission overhead. Still, existing anamorphic signatures are based on symmetric encryption, which requires prior sharing of symmetric keys between the sender and the receiver [22]. The concept of public-key anamorphic proposed by Persiano et al. [23] overcomes this limitation without relying on shared symmetric keys. Nonetheless, Persiano et al. primarily focus on constructing public-key anamorphic encryption schemes rather than signature schemes. Therefore, it does not meet the requirements of our scenario, where a signature scheme is needed to leverage the anamorphic property for optimizing communication overhead.

Given the highly dynamic nature of VANETs, this challenge becomes even more pronounced. In such environments, the computational and communication costs associated with symmetric key pre-sharing are particularly high. Thus, the core problem that needs to be solved in our work emerges:

Can we reformulate anamorphic signatures to achieve compatibility with CL-CPPA for latency-sensitive VANETs while inheriting the special properties of anamorphic signatures?

A. Contributions

We affirm that the answer to the above challenge is positive by refining the anamorphic signature. Concretely, we will achieve it by embedding ciphertext in the metadata of communications. Authentication and secret communication are performed simultaneously to minimize the computation and communication overhead. The main contributions were threefold.

- We first introduce a novel cryptographic primitive, asymmetric anamorphic signatures (AAS), where the anamorphic message is encrypted asymmetrically to solve the complex key agreement issue faced by existing symmetric anamorphic signatures. Subsequently, we propose a variant of the Hohenberger-Waters scheme (short for H-W$^+$) and give the AAS instance based on it to adapt to latency-sensitive VANETs.
- We further present the first CL-CPPA protocol with secure transmission (*Sphinx*), taking advantage of the real-time communication requirements of VANETs. In addition, we establish a security model that rigorously proves the indistinguishability under chosen plaintext attack (IND-CPA) and existential unforgeability against chosen message attack (EUF-CMA) for our proposed scheme.
- Finally, by the combined consideration of theoretical analysis and experimental simulation, our scheme has better performance than state-of-the-art solutions and can be better adapted to latency-sensitive VANETs. The characteristics are as follows.
  Flexible communication: When users want to communicate with the roadside unit (RSU) or other vehicles, they can decide whether to embed sensitive messages in their signatures according to their needs.
  Conditional identity tracking: When there is a malicious user in the system, the trust authority (TA) can track the real identity of the malicious user through its private key.
  High efficiency: Ensures secure message communication without relying on additional encryption after signatures to ensure confidentiality and without the overhead of additional ciphertext transmissions or complex key agreements.

B. Organization

Section II provides an extensive review of related work. Section III introduces cryptographic primitives and system architecture. In Section IV, we introduce the AAS scheme by describing the H-W$^+$ scheme and proving its security. In Section V, we elaborate on our *Sphinx*, including definition and security models, as well as provide proofs of security and probability analysis under the random oracle model (ROM). Section VI analyzes the performance and efficiency of our scheme to verify the feasibility of our *Sphinx*. Finally, Section VII summarizes this paper.

[1] We referred to the classic IBE scheme [15], [16], whose ciphertext length is 2-4 times that of plaintext.

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II. Related Work

Due to the open and dynamic nature of VANETs, they are prone to challenges. To achieve identity verification and ensure the integrity and non-repudiation of messages, it is essential to sign the messages. In addition, encryption of sensitive messages is necessary when transmitting them. The concepts of CPPA and signcryption have received extensive research attention. CPPA can provide conditional anonymous privacy protection while meeting the aforementioned security requirements. The signcryption scheme can ensure message integrity while maintaining message confidentiality.

This section will provide an overview of representative research results in CPPA and signcryption schemes.

A. CPPA

In 2007, Raya et al. [24] first proposed a PKI-based CPPA scheme aimed at solving security and privacy problems in VANETs. Subsequently, numerous scholars have proposed similar PKI-based CPPA schemes [25], [26], [27], [28]. Although such schemes can achieve conditional privacy protection, the high cost of certificate management makes them difficult to be widely used in resource-constrained VANETs devices. To address the aforementioned certificate management challenges, Zhang et al. [29] designed an identity-based CPPA scheme. Subsequently, many scholars also proposed identity-based CPPA schemes [30], [31], [32], but unfortunately, due to the powerful functionality of the key generation center (KGC) in these schemes, most of them face the problem of key custody.

The concept of CL-CPPA has attracted much attention as one of the measures to address key custody. Horng et al. [33] proposed a CL-CPPA scheme based on VANETs and demonstrated its security. Li et al. [34] believed that this scheme could not prevent malicious but passive KGC attacks, and based on this, proposed a new scheme based on bilinear pairs. In addition, many researchers have also proposed various efficient CL-CPPA schemes [35], [36], [37], [38] to effectively mitigate key escrow issues or minimize frequent interactions between entities [10], [11], [39], [40], [41], [42].

Although these solutions successfully balance the needs of privacy protection and efficient tracking, most of them do not consider the confidentiality of messages and default to commonly used encryption algorithms. Their operational efficiency is inevitably limited by the encryption-decryption operations, incurring some latency, which is detrimental for latency-sensitive VANETs.

B. Signcryption

In VANETs, the receiver is required to confirm the identity of the sender while verifying the integrity and non-repudiation of the message. In addition, the sender needs to transmit sensitive messages through encryption to ensure the confidentiality of the message.

Zheng et al. [13] first proposed the concept of signcryption. Signcryption is a novel encryption technology that can simultaneously perform signature and encryption. Compared to traditional methods, the cost of signcryption is lower. Liu et al. [43] subsequently proposed a certificateless signcryption mechanism and sparked a series of CL-SC schemes [44], [45], [46]. Among them, Ali et al. [45] proposed a hybrid CPPA signcryption scheme with the anonymity property. Although these schemes ensure the unforgeability and confidentiality of messages, the openness of VANETs allows any entity to detect the transmission of sensitive messages and requires the sending of additional ciphertext data during transmission. The additional communication overhead incurred is also detrimental to VANETs.

In 2023, Kutylowski et al. [18] proposed the concept of anamorphic signatures, which encode anamorphic information within signatures, making them indistinguishable from ordinary signatures. This is fundamentally different from signcryption schemes, which typically involve signing then encrypting or encrypting then signing. Therefore, anamorphic signatures appear to be effective measures to address the aforementioned additional transmission overhead. However, current anamorphic signatures rely on symmetric cryptosystems (which require the maintenance of a symmetric double key between two entities), making it unsuitable for being applied to CL-CPPA directly for dynamic VANETs.

III. Preliminaries

In this section, we describe the involved symbols, cryptographic primitives, and system framework as follows.

A. Involved Symbols

For ease of reading, Table I summarizes the relevant symbols of our *Sphinx*.

B. Cryptographic Primitives

CL-CPPA: The generic construction of the CL-CPPA scheme has been summarized in [11], and we give a brief description here. It primarily comprises the following eight algorithms.

1) $\mathsf{Setup}(\lambda) \rightarrow (SP, sk_{\mathsf{KGC}}, sk_{\mathsf{TA}})$*:* The algorithm is usually done jointly by KGC and TA to obtain the public parameters to initialize the system. It takes safety parameter 1^λ as input and outputs the public system parameter SP, keeping $(sk_{\mathsf{KGC}}, sk_{\mathsf{TA}})$ in secret.
2) $\mathsf{PseudonymGeneration}(SP, ID_i) \rightarrow PID_i$*:* The algorithm is usually done jointly by users and TA to generate pseudonyms. It takes the public system parameter SP and the real identity ID_i of users as inputs and outputs the pseudonym PID_i.
3) $\mathsf{Traceability}(SP, PID_i, sk_{\mathsf{TA}}) \rightarrow ID_i$*:* This algorithm is used to track the true identity of malicious users and can only be completed by TA. It takes the pseudonym PID_i of users and the private key of the TA (sk_{TA}) as inputs and outputs the true identity ID_i for malicious users.
4) $\mathsf{SetSecretValue}(SP) \rightarrow (x_i, X_i)$*:* The algorithm is run by the user, inputting SP and then outputting (x_i, X_i).
5) $\mathsf{PartialPrivateKeyGeneration}(SP, PID_i, X_i) \rightarrow (b_i, A_i)$*:* KGC executes the algorithm. It takes as

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TABLE I
SYMBOL DESCRIPTION

Notation	Description
$\mathbb{PG}$	public system parameters
$H(\cdot)$	cryptographic hash function
msk	system master key
mpk	KGC's public key
$(sk_{\mathsf{TA}}, pk_{\mathsf{TA}})$	trust authority (TA)'s public-private key
$\mathcal{V}_i, \mathcal{R}_i$	the i-th vehicle/RSU
ID_i, PID_i	$\mathcal{V}_i$'s real identity and pseudonym
$ID_{\mathcal{R}_i}$	$\mathcal{R}_i$'s real identity
$(sk_{\mathcal{V}_i}, pk_{\mathcal{V}_i})$	$\mathcal{V}_i$'s public-private key
$(sk_{\mathcal{R}_i}, pk_{\mathcal{R}_i})$	$\mathcal{R}_i$'s public-private key
b_i	$\mathcal{V}_i$'s partial private key
t_i	the current timestamp
m	a public message
am	a sensitive message that needs to keep secret
r	ciphertext
$asig$	anamorphic signature on message m
PPT	probability polynomial time
$\mathcal{C}_1/\mathcal{C}_2/\mathcal{C}_3$	simulators
$\mathcal{A}_1/\mathcal{A}_2/\mathcal{A}_3$	three types of adversaries
Ψ = (*KG, Enc, Dec*)	a symmetric encryption algorithm
Υ = (*KG, Enc, Dec*)	an asymmetric encryption algorithm
Φ = (*KG, Sig, Ver*)	a conventional signature algorithm

input the public system parameter SP, the pseudonyms PID_i and X_i of the users, and outputs the partial private key (b_i, A_i).

6) $\mathsf{KeyGeneration}(SP, PID_i, A_i) \rightarrow (sk_i, pk_i)$*:* Users execute this algorithm. It inputs the pseudonym PID_i and partial private key A_i and outputs public-private key pairs for users.
7) $\mathsf{Sign}(SP, PID_i, m, sk_i) \rightarrow \sigma$*:* The algorithm is run by the sender, who signs the message m to get σ. Subsequently, the sender broadcasts the message-signature pair (m, σ).
8) $\mathsf{Verify}(SP, PID_i, m, \sigma, pk_i) \rightarrow (0/1)$*:* After receiving the (m, σ), the receiver runs this algorithm. All entities in the system can verify the validity of the signature using this algorithm.

Hohenberger-Waters (H-W) Signature: As described in [47], it contains four algorithms, as follows.

1) $\mathsf{Setup}(\lambda) \rightarrow pp$*:* Input a security parameter λ, choose a pairing group $pp = (\mathbb{G}, \mathbb{G}_T, g, p, e)$, and output public system parameters pp.
2) $\mathsf{KG}(pp) \rightarrow (sk, pk)$*:* Input the public system parameter pp, and output the public-private key pair (sk, pk)*:* It randomly chooses $\alpha \in \mathbb{Z}_p, v_1, v_2, v_3, z_1, z_2 \xleftarrow{\$} \mathbb{G}$, computes $g_1 = g^\alpha$, selects the upper bound on the number of signatures, denoted by N, and returns $pk = (g_1, v_1, v_2, v_3, z_1, z_2, N), sk = (\alpha, c)$, where c is a counter that is initialized to $c = 0$.

Algorithm 1: Anamorphic Signature (aSig).

Input: m, am, ask, dk
Output: $asig$
1: select $m, am \in \mathbb{Z}_p^*$
2: $r = \Psi.\mathsf{Enc}(am, dk)$
3: $\sigma_1 = \Phi.\mathsf{Sig}(m, ask)$
4: $asig = (\sigma_1, r)$
5: **return** $asig$

3) $\mathsf{Sig}(pp, m, pk, sk) \rightarrow \sigma$*:* Input the public system parameters pp, message m, and private key sk, output the signature σ. As details, it chooses $r, s \in_R \mathbb{Z}_p$ and increments the counter by one, i.e., $c := c + 1$. It terminates if $c > N$; Otherwise, $\sigma = (\sigma_1, \sigma_2, \sigma_3, \sigma_4) = ((v_1^m v_2^r v_3)^\alpha (z_1^c z_2)^s, g^s, r, c)$.
4) $\mathsf{Verf}(pp, m, \sigma, pk) \rightarrow (0/1)$*:* Input the public system parameter pp, message signature pair (m, σ), and public key pk. It first parses the $\sigma = (\sigma_1, \sigma_2, \sigma_3, \sigma_4)$. Once $\sigma_4 \leq N$ and $e(\sigma_1, g) = e(v_1^m v_2^{\sigma_3} v_3, g_1) e(z_1^{\sigma_4} z_2, \sigma_2)$, the validity of the signature can be confirmed.

Remark 1: To better suit our *Sphinx*, we refine H-W to get the new signature scheme, H-W$^+$*:* It will be described in detail in Section IV-B1.

Security Model of H-W$^+$ Signature. This game is an interaction between the simulator $\mathcal{C}$ and the adversary $\mathcal{A}$. The game is defined as follows.

1) *Setup:* $\mathcal{C}$ runs H-W$^+$.KG to gain the public system parameters, pp, and public-private key (sk, pk). Subsequently, $\mathcal{C}$ sends the $\{pp, pk\}$ to $\mathcal{A}$, while maintaining sk secret.
2) *Query:* In this phase, $\mathcal{A}$ makes signature queries on m_i adaptively, and $\mathcal{C}$ responds accordingly.
3) *Forgery:* Finally, at the end of the above queries, $\mathcal{A}$ returns a forged signature $asig^*$ in response to the message m^*. If the signature is valid, then $\mathcal{A}$ wins the game; Otherwise, $\mathcal{A}$ loses. Note that $\mathcal{A}$ cannot query the private key during the forgery process.

Definition 1 (EUF-CMA): If there is no PPT adversary who can win the game with a non-negligible advantage, then the scheme achieves existential unforgeability against chosen-message attacks.

Anamorphic Signature: As described in [18], it is a special form of secure communication that can be implemented in strong adversarial environments, where ciphertext can be embedded into the signature for transmission and is usually composed of three algorithms. What is noteworthiness is that existing anamorphic signatures are symmetric, which already defaults to both parties having the double key.

1) $\mathsf{aKG}(1^\lambda) \rightarrow (ask, apk, dk)$*:* Input the security parameter λ, output the public-private key pair (ask, apk) and the double key dk, where dk is used for subsequent symmetric encryption.
2) $\mathsf{aSig}(m, am, ask, dk) \rightarrow asig$*:* Input the public message m, anamorphic message am, the private key ask, and symmetric key dk, output the anamorphic signature $asig$, as shown in Algorithm 1.

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Algorithm 2: Anamorphic Decryption (aDec).

Input: $asig, apk, dk$
Output: am
Parse $asig$ as (σ_1, r)
if reveivers have dk **then**
  $am = \Psi.\text{Dec}(r, dk)$
else
  $(0/1) \leftarrow \Phi.\text{Ver}(asig, apk)$
end if
return am

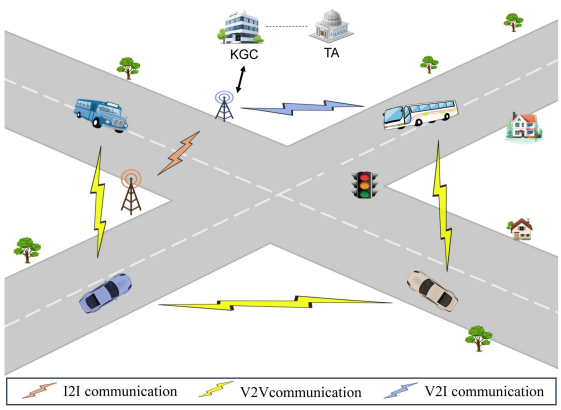

Fig. 1. System model of VANETs.

3) $\mathsf{aDec}(asig, apk, dk) \rightarrow am$: Input the anamorphic signature $asig$, public key apk, and symmetric key dk, return the anamorphic message am, as shown in Algorithm 2.

C. System Framework

Fig. 1 shows in detail the framework and basic communication model of VANETs, which consists of four entities, namely, TA, KGC, RSU ($\mathcal{R}_i$), and Vehicle ($\mathcal{V}_i$). The communication model mainly contains vehicle-to-infrastructure (V2I), vehicle-to-vehicle (V2V), infrastructure-to-infrastructure (I2I), etc. For readability and without loss of generality, this paper focuses on V2I.

TA: TA is completely trustworthy and generates pseudonyms for each $\mathcal{V}_i$. At the same time, to prevent potential misuse or information leakage if a vehicle becomes malicious, the TA retains the capability to trace the real identity of the corresponding vehicle.

KGC: KGC is semi-trustworthy. Traditionally, vehicles require a fully trusted key management center. To enhance the overall security of the system, the trust assumption for vehicle key management has been relaxed from fully trusted to semi-trusted. That is, the system remains secure even under the honest-but-curious model. The main task is to generate a partial private key for $\mathcal{V}_i$. It has powerful computing power and sufficient storage capacity.

$\boldsymbol{\mathcal{R}_i}$*:* RSU is infrastructure equipment installed on both sides of the road. They usually communicate with vehicles through DSRC protocol and require timely verification of the integrity and validity of received messages.

Algorithm 3: Asymmetric Anamorphic Signature (aSig_A).

Input: m, am, sk_s, pk_r
Output: $asig$
select $m, am \in \mathbb{Z}_p^*$
$r = \Upsilon.\text{Enc}(am, pk_r)$
$\sigma_1 = \Phi.\text{Sig}(m, sk_s)$
$asig = (\sigma_1, r)$
return $asig$

Algorithm 4: Asymmetric Anamorphic Decryption (aDec_A).

Input: $asig, pk_s, sk_r$
Output: am
Parse $asig$ as (σ_1, r)
if $1 \leftarrow \Phi.\text{Ver}(asig, pk_s)$ **then**
  $am = \Upsilon.\text{Dec}(r, sk_r)$
else
  return $\perp$
end if
return am

$\boldsymbol{\mathcal{V}_i}$*:* Vehicles are important entities in the system, but they are semi-trusted. They can broadcast real-time traffic situations and communicate with other entities in the system. Each vehicle is equipped with a tamper-proof-device (TPD), which is designed to securely store private keys as well as critical system parameters necessary, but storage capacity and computing power are limited.

IV. Asymmetric Anamorphic Signatures

In this section, we introduce a new cryptographic primitive, the AAS. Specifically, we present the H-W$^+$ scheme and, based on this, provide a concrete construction of the AAS.

A. Definition and Security Model

AAS: Asymmetric anamorphic signature includes the following algorithms.

1) $\mathsf{Setup}(1^\lambda) \rightarrow pp$: Input the security parameter λ, output the public system parameter pp.
2) $\mathsf{aKG}_A^s(pp) \rightarrow (sk_s, pk_s)$: Input the public system parameter pp, output the public-private key pair of senders (sk_s, pk_s).
3) $\mathsf{aKG}_A^r(pp) \rightarrow (sk_r, pk_r)$: Input the public system parameter pp, output the public-private key pair of receivers (sk_r, pk_r).
4) $\mathsf{aSig}_A(m, am, sk_s, pk_r) \rightarrow asig$: Input the public message m, anamorphic message am, the private key of the sender sk_s, and the public key of the receiver pk_r, output the anamorphic signature $asig$, as shown in Algorithm 3.
5) $\mathsf{aDec}_A(asig, pk_s) \rightarrow am$: Input the anamorphic signature $asig$, public key of the sender pk_s, and private key of the receiver sk_r, return the anamorphic message am, as shown in Algorithm 4.

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Furthermore, we need to establish the indistinguishability between two games $(\mathsf{RG}_{\Phi,\mathcal{A}}, \mathsf{AG}_{\mathsf{T},\mathcal{A}})$ to define that an unforgeable signature scheme is asymmetric anamorphic. Here, $\mathsf{RG}_{\Phi,\mathcal{A}}$ denotes the real game, which corresponds to the conventional signature game, while $\mathsf{AG}_{\mathsf{T},\mathcal{A}}$ represents the anamorphic game, that is, the game for anamorphic signatures.

Definition 2 (Asymmetric Anamorphic Signatures): An unforgeable signature scheme $\Phi = (\mathsf{KG}, \mathsf{Sig}, \mathsf{Ver})$ is asymmetric anamorphic if there exists an asymmetric anamorphic triplet $\mathsf{T} = (\mathsf{aKG}^{\mathsf{s}}_{\mathsf{A}}, \mathsf{aKG}^{\mathsf{r}}_{\mathsf{A}}, \mathsf{aSig}_{\mathsf{A}}, \mathsf{aDec}_{\mathsf{A}})$ such that for every PPT adversary $\mathcal{A}$ there exists a negligible function negl such that

$$|\Pr[\mathsf{RG}_{\Phi,\mathcal{A}}(\lambda) = 1] - \Pr[\mathsf{AG}_{\mathsf{T},\mathcal{A}}(\lambda)]| \leqslant \mathsf{negl}(\lambda),$$

where

$\mathsf{RG}_{\Phi,\mathcal{A}}(\lambda)$
1: (pk, sk) $\leftarrow \$\mathsf{KG}(1^\lambda)$
2: **return** $\mathcal{A}^{O_s(\cdot,\cdot,sk)}(pk, sk)$, where
$O_s(m, am, sk) = Sig(m, sk)$.
$\mathsf{AG}_{\mathsf{T},\mathcal{A}}(\lambda)$
1: $(pk_i, sk_i) \leftarrow \$\mathsf{aKG}^{\mathsf{i}}_{\mathsf{A}}(1^\lambda)$
2: **return** $\mathcal{A}^{O_{as}(\cdot,\cdot,sk_s,pk_r)}(pk_s, sk_s)$, where
$O_{as}(m, am, sk_s, pk_r) = aSig(m, am, sk_s, pk_r)$.

Security Model: In fact, the primary objective of the AAS scheme is to conceal anamorphic information. Specifically, the AAS scheme reflects indistinguishability through the portrayal of the game $\mathsf{aCPAG}^{\eta}_{\mathsf{T},\mathcal{A}}$, for an asymmetric anamorphic 4-tuple $\mathsf{T} = (\mathsf{aKG}^{\mathsf{s}}_{\mathsf{A}}, \mathsf{aKG}^{\mathsf{r}}_{\mathsf{A}}, \mathsf{aSig}_{\mathsf{A}}, \mathsf{aDec}_{\mathsf{A}})$, the challenge $\mathcal{C}$, PPT adversary $\mathcal{A}$, and bit $\eta \in \{0, 1\}$.

Definition 3 (IND-CPA): Asymmetric anamorphic scheme AAS is IND-CPA secure if for all PPT adversary $\mathcal{A}$ we have

$$|\Pr[\mathsf{aCPAG}^0_{\mathsf{T},\mathcal{A}}(\lambda) = 1] - \Pr[\mathsf{aCPAG}^1_{\mathsf{T},\mathcal{A}}(\lambda) = 1]| \leqslant \mathsf{negl}(\lambda),$$

where

$\mathsf{aCPAG}^{\eta}_{\mathsf{T},\mathcal{A}}(\lambda)$
1: $(pk_s, sk_s) \leftarrow \mathsf{aKG}^{\mathsf{s}}_{\mathsf{A}}(1^\lambda)$
2: $(pk_r, sk_r) \leftarrow \mathsf{aKG}^{\mathsf{r}}_{\mathsf{A}}(1^\lambda)$
3: $(m, am_0, am_1) \leftarrow \mathcal{A}^{O_e(\cdot,\cdot,sk_s,pk_r)}(pk_s, sk_s)$
4: $asig = O^{\eta}_c(m, am_0, am_1, sk_s, pk_r)$
5: **return** $\mathcal{A}^{O_e(\cdot,\cdot,sk_s,pk_r)}(asig)$, where
$O_e(m, am, sk_s, pk_r) = aSig(m, am, sk_s, pk_r)$
$O^{\eta}_c(m, am_0, am_1, sk_s, pk_r) = aSig(m, am_\eta, sk_s, pk_r)$.

B. Construction From Hohenberger-Waters$^+$

To implement a public-key anamorphic signature, if a portion of the signature (such as $\sigma_3 = r$) is the ciphertext, then another random number s needs to be introduced for encryption. Ensure that s is not compromised (such as $\sigma_2 = g^s$), and the composition of the signature must contain at least three parts, such as $\sigma = (\sigma_1, \sigma_2, \sigma_3)$. Furthermore, embedding the sensitive message in σ_1 or σ_2 is not feasible, as these components cannot carry ciphertext without breaking their structure. Therefore, we embed the sensitive message only in the randomness part of σ_3, which preserves the appearance and security of a traditional signature.

We have found that the Hohenberger-Waters [47] scheme may be suitable, and its security under the CDH assumption has been demonstrated. However, the signature of the H-W program consists of four parts, whereas in our vision, it does not need to be so complex. To better align with our ideas, we are attempting to refine it.

1) Hohenberger-Waters$^+$ Scheme: The H-W$^+$ scheme is as shown in Fig. 2.

2) Asymmetric Anamorphic Signatures: Based on the construction of the H-W$^+$ scheme, we then propose a specific construction of the AAS, as shown in Fig. 3.

It is worth noting that the definition of AAS in our work is fundamentally different from the public-key anamorphic encryption scheme AME proposed at CRYPTO'24 [23]. Specifically, AME is defined as a triple $\mathsf{T} = (\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$, where aEnc outputs a ciphertext act. In contrast, in our AAS construction, aSig outputs a signature $asig$. This is the primary distinction between the two schemes.

C. Security Proof

In this subsection, we prove the security of H-W$^+$ and AAS. The correctness of the scheme is similar to that of the original H-W scheme, and it only replaces $z_1^c z_2$ with g_2 for calculation. The security of H-W$^+$ is shown as follows.

Theorem 1: The H-W$^+$ scheme is secure under the EUF-CMA assuming the hardness of the CDH problem.

Remark 2: The proof of Theorem 1 is given in the Appendix, available online.

Theorem 2: Let AAS be an asymmetric anamorphic signature scheme, and then AAS is secure in the IND-CPA model.

Proof: We prove its security by proving a contradiction, assuming that there exists a PPT adversary $\mathcal{A}$ whose probability of distinguishing the ciphertext is non-negligible, such that

$$\Pr[\mathrm{aCPAG}^1_{\mathrm{T},\mathcal{A}}(\lambda) = 1] \geq \Pr[\mathrm{aCPAG}^0_{\mathrm{T},\mathcal{A}}(\lambda) = 1] + \frac{1}{\mathrm{poly}(\lambda)}.$$

The detailed proof process is similar to CRYPTO'23 [18] and will not be repeated here, where the difference with CRYPTO'23 is that the symmetric key of the encryption process is replaced by the public key of the receiver. □

V. Our Sphinx

In this section, we describe a formal definition and security models for *Sphinx*, followed by describing its concrete construction, which was proved secure.

A. Definition and Security Model

Sphinx: In our work, the *Sphinx* includes the following algorithms.

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KG(λ)	Sig(pp, m, pk, sk)	Verf(pp, m, σ, pk)
1. $pp = (\mathbb{G}, \mathbb{G}_T, g, p, e)$	1. $r, s \xleftarrow{\$} \mathbb{Z}_p^*$	1. Parse $\sigma = (\sigma_1, \sigma_2, \sigma_3)$
2. $\alpha \xleftarrow{\$} \mathbb{Z}_p^*, g_1 = g^\alpha$	2. $\sigma_1 = (v_1^m v_2^r v_3)^\alpha \cdot g_2^s$	2. Parse $pk = (g_1, v_1, v_2, v_3, g_2)$
3. $v_1, v_2, v_3, g_2 \xleftarrow{\$} \mathbb{G}$	3. $\sigma_2 = g^s$	3. If $e(\sigma_1, g) = e(v_1^m v_2^{\sigma_3} v_3, g_1) e(g_2, \sigma_2)$,
4. $sk = \alpha$	4. $\sigma_3 = r$	4. return 1;
5. $pk = (g_1, v_1, v_2, v_3, g_2)$	5. $\sigma = (\sigma_1, \sigma_2, \sigma_3)$	5. Otherwise,
6. retun (sk, pk)	6. return (m, σ)	6. return 0

Fig. 2. The hohenberger-waters$^+$ scheme.

Setup(λ)

1. Choose a pairing group $\mathbb{PG} = (\mathbb{G}, \mathbb{G}_T, g, p, e)$
2: $H : \{0,1\}^* \to \mathbb{Z}_p^*$
3. $pp = (\mathbb{PG}, H)$
4. retun pp

aKG$_A^s(pp)$

1. $\alpha \xleftarrow{\$} \mathbb{Z}_p^*, g_1 = g^\alpha$
2. $v_1, v_2, v_3, g_2 \xleftarrow{\$} \mathbb{G}$
3. $sk_s = \alpha, pk_s = (g_1, v_1, v_2, v_3, g_2)$
4. return (sk_s, pk_s)

aKG$_A^r(pp)$

1. $\beta \xleftarrow{\$} \mathbb{Z}_p^*$
2. $sk_r = \beta$
3. $pk_r = g^\beta$
4. return (sk_r, pk_r)

aSig$_A(pp, m, am, pk_s, sk_s, pk_r)$

1: $s \xleftarrow{\$} \mathbb{Z}_p^*$
2: $m, am \in \mathbb{Z}_p^*$
3: $h = H(m, g^s)$
4: $r = H(pk_r^s) \oplus am$
5: $\sigma_1 = (v_1^h v_2^r v_3)^{sk_s} \cdot g_2^s$
6. $\sigma_2 = g^s$
7. $\sigma_3 = r$
8. $asig = (\sigma_1, \sigma_2, \sigma_3)$
9: return $(m, asig)$

aDec$_A(pp, m, asig, pk_s, sk_r)$

2: Parse pk_s as $(g_1, v_1, v_2, v_3, g_2)$
3: Parse $asig$ as $(\sigma_1, \sigma_2, \sigma_3)$
5: $h = H(m, g^s)$
6: If $e(\sigma_1, g) \neq e(v_1^h v_2^{\sigma_3} v_3, g_1) e(g_2, \sigma_2)$, abort
7: $am = r \oplus H(\sigma_2^{sk_r})$
8: return am

Fig. 3. The asymmetric anamorphic signature scheme.

1) The definitions of the algorithms $\mathsf{Setup}, \mathsf{U-PID}, \mathsf{Trace}, \mathsf{U-Secret}, \mathsf{U-PPK}, \mathsf{U-KG}$ are similar to the definition of CL-CPPA in Section III-B.
2) $\mathsf{R-KG}$*:* This algorithm is completed by TA, generating the public-private key pair for RSU.
3) $\mathsf{A-Sign}$*:* This algorithm selects two types of messages, which are public message, m, and sensitive message, am. The sender signs m to obtain σ_1, and encrypts am to obtain r as another part of the signature. Finally, output the full signature $asig$.
4) $\mathsf{Ver-Dec}$*:* After receiving the $asig$, the receiver runs this algorithm. All receivers can verify the validity of the signature using this algorithm, but only the specific receiver (those who want to communicate secretly) can recover the sensitive message am using this algorithm.

More specifically, as depicted in Fig. 4, the application of the *Sphinx* system model to VANETs can be succinctly summarized as follows.

① *Registration:* This stage is mainly divided into three parts.

a. KGC and TA run Setup to get system parameter SP and private key (msk, sk_{TA}) respectively. Keep (msk, sk_{TA}) secretly and release SP publicly;
b. KGC runs $\mathsf{U-PPK}$ to obtain the partial private key (b_i, A_i), and TA runs $\mathsf{U-PID}$ to generate PID_i;
c. $\mathcal{V}_i$ runs $\mathsf{U-Secret}, \mathsf{U-KG}$ to generate $(X_i, sk_{\mathcal{V}_i}, pk_{\mathcal{V}_i})$, and $\mathcal{R}_i$ runs $\mathsf{R-KG}$ to gain $(sk_{\mathcal{R}_i}, pk_{\mathcal{R}_i})$.

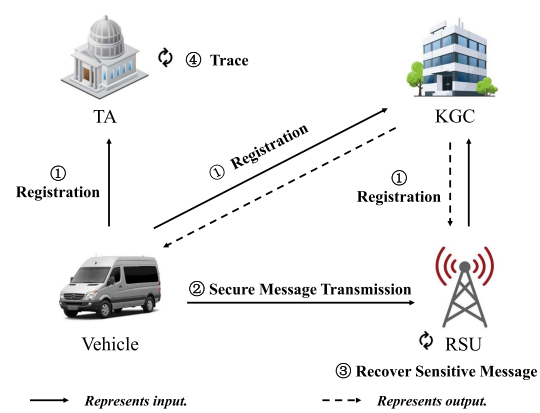

Fig. 4. System structure diagram.

② *Secure Message Transmission:* At this stage, when $\mathcal{V}_i$ wants to send a sensitive message to $\mathcal{R}_i$, it runs $\mathsf{A-Sign}$ to generate a signature and broadcasts it.

③ *Recover Sensitive Message:* At this stage, all receivers can run $\mathsf{Ver-Dec}$ to verify the signature (enabling public verification). However, only the $\mathcal{R}_i$ that $\mathcal{V}_i$ intends to communicate with secretly can use $\mathsf{Ver-Dec}$ to decrypt the hidden sensitive message in the signature.

④ *Trace:* TA runs Trace to trace the true identity of malicious users and revoke it, which can only be operated by TA.

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Security Models: We consider that the scheme in VANETs should at least satisfy the requirements of confidentiality and unforgeability, i.e., IND-CPA and EUF-CMA.

Sphinx aims to hide sensitive messages and enable secure transmission. This, for adversaries $\mathcal{A}_1$, not only suggests that the ciphertexts are indistinguishable but also implies that the presence or absence of anamorphic message is equally indeterminable. Furthermore, our *Sphinx* is based on certificateless to achieve unforgeability. Therefore, we consider the following two types of adversaries based on different malicious behaviors: $\mathcal{A}_2$ and $\mathcal{A}_3$.

Before detailing these games, first define some common oracle queries as follows.

1) O_{H}: Adversaries perform hash queries, and this oracle returns the hash of messages. When $\mathcal{A}$ inputs $m \in \{0,1\}^*$, the simulator $\mathcal{C}$ returns the according value to the list $\mathcal{L}$.
2) O_{PK}: This oracle returns the $\mathcal{V}_i$'s partial private key. $\mathcal{A}$ queries the $\mathcal{V}_i$'s partial private key from $\mathcal{C}$, $\mathcal{C}$ returns (b_i, A_i). q_{pk} denotes the number of queries to the oracle.
3) $O_{\mathsf{U.K.}}$: This oracle returns the $\mathcal{V}_i$'s public key. $\mathcal{A}$ queries $\mathcal{C}$ for the public key of $\mathcal{V}_i$, and $\mathcal{C}$ returns $pk_{\mathcal{V}_i}$ to $\mathcal{A}$. Additionally, we use q_{UK} to represent the number of queries for the oracle.
4) O_{PKR}: This oracle can replace the public key. $\mathcal{A}$ replaces the public key and sends a new public key to $\mathcal{C}$. q_{pkr} represents the number of queries.
5) O_{USK}: This oracle returns the $\mathcal{V}_i$'s private key. $\mathcal{A}$ queries the private key of $\mathcal{V}_i$, and $\mathcal{C}$ returns $sk_{\mathcal{V}_i}$ to $\mathcal{A}$. q_{usk} denotes the number of queries to the oracle.
6) O_{RK}: It returns the $\mathcal{R}_i$'s public key. $\mathcal{A}$ queries the simulator $\mathcal{C}$ for the public key of $\mathcal{R}_i$, and $\mathcal{C}$ returns $pk_{\mathcal{R}_i}$ to $\mathcal{A}$. q_{rk} represents the number of queries.
7) O_{S}: It returns the result of the signature. $\mathcal{A}$ queries the signature of message, and $\mathcal{C}$ returns $(m, asig)$ to $\mathcal{A}$. q_s to represent the number of signature queries.
8) O_{E}: It returns the result of encryption. $\mathcal{A}$ sends plaintext am to $\mathcal{C}$, $\mathcal{C}$ calculates the ciphertext ct via $\mathsf{A-Sign}$ algorithm and returns it to $\mathcal{A}$.

$\mathbf{Game_1^\eta}$: This game is an interaction between the simulator $\mathcal{C}_1$ and the adversary $\mathcal{A}_1$ under IND-CPA. The game is defined as follows.

1) *Initialization:* $\mathcal{C}_1$ runs Setup to gain the public system parameters, $SP = \{\mathbb{PG}, H, mpk, pk_{\mathsf{TA}}\}$, and private key (msk, sk_{TA}). Subsequently, $\mathcal{C}_1$ sends the SP to $\mathcal{A}_1$, while maintaining msk, sk_{TA} secret.
2) *Query:* In this phase, $\mathcal{A}_1$ makes a series of queries $O_{\mathsf{H}}, O_{\mathsf{PK}}, O_{\mathsf{U.K.}}, O_{\mathsf{RK}}, O_{\mathsf{E}}$ adaptively, and $\mathcal{C}_1$ responds accordingly.
3) *Challenge:* After the above queries, $\mathcal{A}_1$ selects two plaintext $\{am_0, am_1\}$. Then, $\mathcal{C}_1$ randomly selects $\eta \in \{0,1\}$ and computes ct^*. Finally, $\mathcal{C}_1$ sends ct^* to $\mathcal{A}_1$.
4) *Guess:* In this phase, the $\mathcal{A}_1$ outputs a bit $\eta' \in \{0,1\}$. If $\eta' = \eta$, $\mathcal{A}_1$ wins the game; Otherwise, $\mathcal{A}_1$ fails.

Definition 4 (IND-CPA): As long as there is no PPT adversary $\mathcal{A}_1$ who can win $\mathbf{Game_1^\eta}$ with a non-negligible advantage, the scheme is secure under IND-CPA.

$\mathbf{Game_2}$: This game is an interaction between the simulator $\mathcal{C}_2$ and the adversary $\mathcal{A}_2$ under EUF-CMA, where $\mathcal{A}_2$ simulates an external attacker, i.e., the public key can be replaced. The game is defined as follows.

1) *Setup:* Adversary $\mathcal{A}_2$ and $\mathcal{C}_2$ perform the same operations as in $\mathbf{Game_1^\eta}$.
2) *Query:* In this phase, $\mathcal{A}_2$ makes a series of queries $O_{\mathsf{H}}, O_{\mathsf{PK}}, O_{\mathsf{U.K.}}, O_{\mathsf{RK}}, O_{\mathsf{S}}$ adaptively, and $\mathcal{C}_2$ responds accordingly.
3) *Forgery:* Finally, at the end of the above queries, $\mathcal{A}_2$ returns a forged signature $asig^*$ in response to the message m^*. If the signature is valid, then $\mathcal{A}_2$ wins the game; Otherwise, $\mathcal{A}_2$ loses. Note that $\mathcal{A}_2$ cannot query the private key during the forgery process.

$\mathbf{Game_3}$: This game is an interaction between the simulator $\mathcal{C}_3$ and the adversary $\mathcal{A}_3$ under EUF-CMA, where $\mathcal{A}_3$ simulates an internal attacker, i.e., obtains a part of the private key. The game is defined as follows.

1) *Setup:* Adversary $\mathcal{A}_3$ and $\mathcal{C}_3$ perform the same operations as in $\mathbf{Game_2}$.
2) *Query:* In this phase, $\mathcal{A}_3$ makes a series of queries $O_{\mathsf{H}}, O_{\mathsf{PK}}, O_{\mathsf{U.K.}}, O_{\mathsf{RK}}, O_{\mathsf{S}}$ adaptively, and $\mathcal{C}_3$ responds accordingly.
3) *Forgery:* Finally, at the end of the above queries, $\mathcal{A}_3$ returns a forged signature $asig^*$ in response to the message m^*. If the signature is valid, then $\mathcal{A}_3$ wins the game; Otherwise, $\mathcal{A}_3$ loses. Note that $\mathcal{A}_3$ cannot query the private key during the forgery process.

Definition 5 (EUF-CMA): As long as there are no PPT adversaries ($\mathcal{A}_2$ and $\mathcal{A}_3$) who can win $\mathbf{Game_2}$ and $\mathbf{Game_3}$ with a non-negligible advantage, the Sphinx is secure under EUF-CMA.

Security Objectives: Taking into the variety of real-world attacks, a viable *Sphinx* suitable for VANETs must satisfy the following fundamental security requirements.

1) *Message Authentication:* Any entity in the system can verify the authenticity and integrity of information from other vehicles. This means that any modifications made to the messages will be detected.
2) *Confidentiality:* The receiver cannot distinguish the ciphertext, and other users in the system cannot recognize whether sensitive messages are embedded in the signature. This means that sensitive messages that need to be kept confidential can be transmitted silently without being discovered by other entities.
3) *Non-Repudiation:* It is impossible for the signer of any messages to refute the authenticity of the signature within the system, effectively thwarting attackers' attempts to deny or conceal illicit activities.
4) *Identity Anonymity:* In the process of identity authentication, the scheme should ensure the privacy of the user's true identity so that no other users (or Adversaries) can obtain the user's true identity.
5) *Condition Tracking:* Provides that only the TA can track the identity of malicious users and that other entities are not qualified to recover the true identity; in other words, the TA can regulate users.

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Setup(1^λ)

1: Choose a pairing group $\mathbb{PG} = (\mathbb{G}, \mathbb{G}_T, g, p, e)$
2: $H : \{0,1\}^* \to \mathbb{Z}_p^*$
3: $msk \xleftarrow{\$} \mathbb{Z}_p^*, mpk \leftarrow g^{msk}$
4: $sk_{\mathsf{TA}} \xleftarrow{\$} \mathbb{Z}_p^*, pk_{\mathsf{TA}} \leftarrow g^{sk_{\mathsf{TA}}}$
5: return $SP = \{\mathbb{PG}, H, mpk, pk_{\mathsf{TA}}\}$

U-PID(SP, ID_i)

1: $k \xleftarrow{\$} \mathbb{Z}_p^*, PID_{i,1} = g^k$
2: $PID_{i,2} = H(pk_{\mathsf{TA}}^k) \oplus ID_i$
3: return $PID_i = \{PID_{i,1}, PID_{i,2}\}$

U-Secret(SP, PID_i)

1: $x_i \xleftarrow{\$} \mathbb{Z}_p^*, X_i \leftarrow g^{x_i}$
2: return (x_i, X_i)

U-PPK(SP, PID_i, X_i)

1: $a_i \xleftarrow{\$} \mathbb{Z}_p^*, A_i \leftarrow g^{a_i}$
2: $h_1 \leftarrow H(PID_i||X_i||A_i)$
3: $b_i = a_i - h_1 \cdot msk$
4: return (b_i, A_i)

U-KG$(SP, PID_i, X_i, h_1, b_i)$

1: $u_1, u_2, u_3, v \xleftarrow{\$} \mathbb{G}$
2: $\kappa = b_i + x_i$
3: $sk_{\mathcal{V}_i} = \kappa$
4: $pk_{\mathcal{V}_i} = (g^\kappa, u_1, u_2, u_3, v)$
5: return $(sk_{\mathcal{V}_i}, pk_{\mathcal{V}_i})$

R-KG$(SP, ID_{\mathcal{R}_i})$

1: $\beta \xleftarrow{\$} \mathbb{Z}_p^*$
2: $(sk_{\mathcal{R}_i}, pk_{\mathcal{R}_i}) = (\beta, g^\beta)$
3: return $(sk_{\mathcal{R}_i}, pk_{\mathcal{R}_i})$

A-Sign$(SP, PID_i, m, am, sk_{\mathcal{V}_i}, pk_{\mathcal{R}_i})$

1: $s \xleftarrow{\$} \mathbb{Z}_p^*$
2: $m \in \{0,1\}^*, am \in \mathbb{Z}_p^*$
3: $h_3 \leftarrow H(m||PID_i||g^s||t_i)$
4: $r = H((pk_{\mathcal{R}_i})^s) \oplus am$
5: $\sigma_1 \leftarrow (u_1^{h_3} u_2^r u_3)^\kappa \cdot v^s, \sigma_2 \leftarrow g^s, \sigma_3 = r$
6: $asig = (\sigma_1, \sigma_2, \sigma_3)$
7: return $(m, asig)$

Ver-Dec$(SP, PID_i, m, asig, t_i, A_i, X_i, pk_{\mathcal{V}_i}, sk_{\mathcal{R}_i})$

1: If $|t_i^{'} - t_i| > \Delta t$, abort
2: Parse $pk_{\mathcal{V}_i}$ as $(g^\kappa, u_1, u_2, u_3, v)$
3: Parse $asig$ as $(\sigma_1, \sigma_2, \sigma_3)$
4: $h_1 \leftarrow H(PID_i||X_i||A_i)$
5: $h_3 \leftarrow H(m||PID_i||g^s||t_i)$
6: If $e(\sigma_1, g) \neq e(u_1^{h_3} u_2^r u_3, A_i X_i \cdot mpk^{-h_1}) e(v, \sigma_2)$, abort
7: $am = r \oplus H(\sigma_2^{sk_{\mathcal{R}_i}})$
8: return am

Trace(SP, PID_i, msk)

1: Parse $PID_i = (PID_{i,2}, PID_{i,1})$
2: $ID_i = PID_{i,2} \oplus H(PID_{i,1}^{sk_{\mathsf{TA}}})$
3: return ID_i

Fig. 5. Our construction of the sphinx.

6) *Unlinkability:* To prevent attackers from tracing the historical path of vehicles, our scheme must guarantee that messages sent by the same vehicle cannot be linked together.

B. Construction

As the features of the AAS scheme are not powerful enough for latency-sensitive VANETs (e.g., it does not allow for conditional anonymity). We have refined the CL-CPPA protocol with secure transmission.

This section elaborates on the *Sphinx* in detail and provides its specific application in VANETs. Specifically, our scheme is shown in Fig. 5, and the detailed algorithm is described below.

1) Setup*:* It takes safety parameter 1^λ as input and outputs the public system parameter SP.
   a. KGC picks a pairing group $\mathbb{PG} = (\mathbb{G}, \mathbb{G}_T, g, p, e)$ and one hash function $H : \{0,1\}^* \to \mathbb{Z}_p^*$.
   b. KGC selects $msk \in \mathbb{Z}_p^*$ as the master private key and sets the public key via $mpk = g^{msk}$.
   c. TA selects $sk_{\mathsf{TA}} \in \mathbb{Z}_p^*$ as the private key, computes $pk_{\mathsf{TA}} = g^{sk_{\mathsf{TA}}}$ as its public key, and keeps sk_{TA} secretly.
   d. Finally, KGC keeps msk secret and discloses the system parameters $SP = \{\mathbb{PG}, H, mpk, pk_{\mathsf{TA}}\}$.
2) $\mathsf{U-PID}$*:* Before communication, each vehicle $\mathcal{V}_i$ must provide its true identity ID_i to TA for registration. Subsequently, TA selects a number $k \in \mathbb{Z}_p^*$ randomly and computes $PID_{i,1} = g^k, PID_{i,2} = H(pk_{\mathsf{TA}}^k) \oplus ID_i$. Finally, TA generates a pseudonym PID_i by encrypting its real identity using pk_{TA}. It is worth noting that the length of ID_i and $H(pk_{\mathsf{TA}}^k)$ should be consistent.
3) Trace*:* If the vehicle engages in inappropriate behavior, TA can parse malicious pseudonyms to obtain $(PID_{i,1}, PID_{i,2})$ and decrypt it using sk_{TA} to restore their true identity. Subsequently, punishment can be imposed.
4) $\mathsf{U-Secret}$*:* Based on PID_i, $\mathcal{V}_i$ randomly selects the secret value x_i and stores it secretly for generating a complete key in the future. Then $\mathcal{V}_i$ calculates $X_i = g^{x_i}$ and finally returns (x_i, X_i).
5) $\mathsf{U-PPK}$*:* $\mathcal{V}_i$ first sends its PID_i, X_i to KGC, then KGC will generate part of its private key through the following steps.
   a. Choose $a_i \in \mathbb{Z}_p^*$ and get A_i via g^{a_i}.
   b. Hash PID, X_i, A_i to obtain h_1.
   c. Obtain b_i via $a_i - h_1 \cdot msk$.
   d. Finally, return (b_i, A_i), where (b_i, A_i) are used as the partial private key of $\mathcal{V}_i$.
6) $\mathsf{U-KG}$*:* After receiving the partial private key, $\mathcal{V}_i$ first needs to verify its validity if and only if all of the following equations are satisfied simultaneously: ① $A_i^v = g^{b_i} mpk^{h_1}$, ② $h_1^v = H(PID_i||X_i||A_i^v)$, ③ $h_1^v = h_1$. Then, $\mathcal{V}_i$ generates its public/private key as follows.
   a. Choose u_1, u_2, u_3, v from $\mathbb{G}$.
   b. Then, obtain the first part of $\mathcal{V}_i$'s private key $\kappa = b_i + x_i$.
   c. Finally, return $sk_{\mathcal{V}_i} = \kappa$ and $pk_{\mathcal{V}_i} = (g^\kappa, u_1, u_2, u_3, v)$, where $sk_{\mathcal{V}_i}$ is kept secret.

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7) $\mathsf{R-KG}$: $\mathcal{R}_i$ sends its real identity $ID_{\mathcal{R}_i}$ to TA. Then, TA chooses β as its private key and obtains the public key via g^{β}.
8) $\mathsf{A-Sign}$: $\mathcal{V}_i$ encrypts sensitive message am and generates the signature of m as follows.
   a. First, $\mathcal{V}_i$ selects sensitive message $am \in \mathbb{Z}_p^*$ and public message $m \in \{0,1\}^*$.
   b. Choose $s \in \mathbb{Z}_p^*$ and t_i, and obtain $\sigma_2 = g^s$.
   c. Calculate the hash of (m, PID_i, g^s, t_i).
   d. Encrypt sensitive messages by fixing the receiver's public key $pk_{\mathcal{R}_i}$ and random value s.
   e. Obtain the signature σ_1 via $\sigma_1 = (u_1^{h_3} u_2^r u_3)^{\kappa} \cdot v^s$, and in addition, $\sigma_3 = r$.
   f. Return the final $asig = (\sigma_1, \sigma_2, \sigma_3)$.
9) $\mathsf{Ver-Dec}$: After the RSU receives a message-signature pair, it first checks $|t_i' - t_i| \leqslant \Delta t$, if it holds, calculates $h_1 = H(PID_i||X_i||A_i), h_3 = H(m||PID_i||\sigma_2||t_i)$, and then it could select whether to verify the signature or recover the anamorphic message based on the user's legitimacy. In detail, running this algorithm will produce two results:
   a. If users are not legitimate receivers (or *Adversaries*) and cannot decrypt to obtain am, they can only verify the validity of conventional message m, meanwhile, the algorithm $\mathsf{Ver-Dec}$ will return "$\perp$".
   The following formula can ensure the correctness of the verification.

$$\begin{aligned} & e(\sigma_1, g) \\ & = e((u_1^{h_3} u_2^r u_3)^{\kappa} \cdot v^s, g) \\ & = e(u_1^{h_3} u_2^r u_3, g^{a_i + x_i - msk \cdot h_i}) e(v, \sigma_2) \\ & = e(u_1^{h_3} u_2^r u_3, A_i X_i \cdot mpk^{-h_1}) e(v, \sigma_2). \end{aligned}$$

   b. If the RSU is a legitimate communicator, RSU can verify signatures while also decrypting to obtain anamorphic messages. Specifically, RSU_i could decrypt the signature and obtain the anamorphic message am via $\sigma_3 \oplus H(\sigma_2^{sk_{\mathcal{R}_i}})$.

C. Security Proof and Analysis

In this subsection, we discuss the security that can be satisfied by our proposal. The details are given as follows.

Security Proof: In this subsection, we will analyze security proofs to verify the security of our *Sphinx*.

Theorem 3: The *Sphinx* is security under the IND-CPA model assuming the ROM and hardness of the CDH problem.

Proof of Theorem 3: Assuming that the adversary $\mathcal{A}_1$ can break the *Sphinx* with a random CDH instance: given $\{g, g^a, g^b\}$, the simulator $\mathcal{C}_1$'s goal is to obtain a solution g^{ab} of CDH through interactions with $\mathcal{A}_1$.

Initialization: $\mathcal{C}_1$ initializes the system parameters $SP = \{\mathbb{PG}, H, mpk, pk_{\mathsf{TA}}\}$ by executing the Setup algorithm and sends SP to $\mathcal{A}_1$.

Queries: $\mathcal{C}_1$ maintains an initially empty list $\mathcal{L} = (PID_i, \psi_i, A_i, \varsigma_i, pk_{\mathcal{V}_i}, \kappa, pk_{\mathcal{R}_i})$ and serves the following oracle queries to $\mathcal{A}_1$, specifically,

- O_{H}: The simulator $\mathcal{C}_1$ maintains two hash lists $\mathcal{L}_{H_1}$: (PID_i, X_i, h_1), $\mathcal{L}_{H_2}$: $(PID_i, m_i, \sigma_2, h_3)$ to record all queries and responses, which is initially empty.
  When $\mathcal{A}_1$ quires (PID_i, X_i) for the first time, if there is a record in $\mathcal{L}_{H_1}$, $\mathcal{C}_1$ returns h_1 as a response; otherwise, $\mathcal{C}_1$ randomly selects χ_i as the response and adds this item to $\mathcal{L}_{H_1}$.
  When $\mathcal{A}_1$ quires (PID_i, m_i, σ_2) for the first time if there is a record in $\mathcal{L}_{H_2}$, $\mathcal{C}_1$ returns h_3 as a response; otherwise, $\mathcal{C}_1$ randomly selects ι_i as the response and adds this item to $\mathcal{L}_{H_2}$.
- O_{PK}: $\mathcal{A}_1$ inquires about the partial private key of PID_i. $\mathcal{C}_1$ checks if PID_i exists in the list $\mathcal{L}$, and if so, returns the tuple (PID_i, b_i, A_i) to $\mathcal{A}_1$. Otherwise, $\mathcal{C}_1$ picks three random numbers $\varphi_i, \chi_i, \psi_i \in_R \mathbb{Z}_p^*$ and sets $a_i = \varphi_i$, $b_i = \psi_i$, $H(PID_i||X_i) = h_1 = \chi_i$. Then it computes $A_i = \psi_i \cdot G - \chi_i \cdot mpk$, where (ψ_i, A_i) is the partial private key. $\mathcal{C}_1$ sends (ψ_i, A_i) to $\mathcal{A}_1$ and adds the tuple $(PID_i, \psi_i, A_i, *, *, *, *)$ to $\mathcal{L}$.
- $O_{\mathsf{U.K.}}$: When $\mathcal{A}_1$ requests the pseudonym PID_i, $\mathcal{C}_1$ checks the list $\mathcal{L}$ and returns the $pk_{\mathcal{V}_i}$ associated with PID_i to $\mathcal{A}_1$. Otherwise, select a random value $\varsigma_i \in_R \mathbb{Z}_p^*$ and compute $X_i = g_i^{\varsigma}$. Finally, $\mathcal{C}_1$ sets $pk_{\mathcal{V}_i} = g^{\psi_i} \cdot X_i$ as the public key and transmits it to $\mathcal{A}_1$. Additionally, $\mathcal{C}_1$ updates $(PID_i, \psi_i, A_i, \varsigma_i, pk_{\mathcal{V}_i}, *, *)$ in the $\mathcal{L}$ list.
- O_{USK}: $\mathcal{A}_1$ makes queries to $\mathcal{C}_1$. $\mathcal{C}_1$ computes $\kappa = \varsigma_i + \psi_i$ as private key, then updates the list $\mathcal{L} = (PID_i, \psi_i, A_i, \varsigma_i, pk_{\mathcal{V}_i}, \kappa, *)$.
- O_{RK}: $\mathcal{A}_1$ queries $pk_{\mathcal{R}_i}$, $\mathcal{C}_1$ sets the public key as $pk_{\mathcal{R}_i} = g^a$, private key $\beta = a$, and returns $pk_{\mathcal{R}_i}$ to $\mathcal{A}_1$. Then updates the list $\mathcal{L} = (PID_i, \psi_i, A_i, \varsigma_i, pk_{\mathcal{V}_i}, \kappa, pk_{\mathcal{R}_i})$.
- O_{E}: $\mathcal{A}_1$ queries the ciphertext ct of am, $\mathcal{C}_1$ computes ct via $\mathsf{A-Sign}$, and returns ct to $\mathcal{A}_1$.

Challenge: Second, $\mathcal{A}_1$ selects messages (am_0, am_1) and returns them to $\mathcal{C}_1$. After receiving, $\mathcal{C}_1$ randomly selects $\eta \in \{0,1\}$ and sets $ct^* = (g^b, H(pk_{\mathcal{R}_i}^b) \oplus am_{\eta})$ while returning ct^* to $\mathcal{A}_1$.

Guess: Finally, $\mathcal{A}_1$ receives ct^*, and according to the previous oracle inquiry, $\mathcal{A}_1$ returns a bit η', and $\mathcal{C}_1$ outputs 1 iff $\eta = \eta'$. Next, $\mathcal{C}_1$ can utilize the capabilities of adversaries to solve the CDH problem as $pk_{\mathcal{R}_i}^b = (g^a)^b = g^{ab}$.

Therefore, we can conclude that $\mathcal{C}_1$ violates the hardness of the CDH problem, so our *Sphinx* is IND-CPA secure. □

Theorem 4: Our *Sphinx* can satisfy EUF-CMA, and simultaneously resist attacks from two types of adversaries, $\mathcal{A}_2$ and $\mathcal{A}_3$.

The proof of Theorem 4 can be achieved through the following two lemmas. Lemma 1 proves that the scheme proposed in this article can resist attacks from Type I adversary $\mathcal{A}_2$; Lemma 2 proves that it can resist attacks from Type II adversary $\mathcal{A}_3$. The proof of Lemmas 1 and 2 are given in the Appendix, available online.

Lemma 1: In the ROM, suppose that there is a PPT adversary $\mathcal{A}_2$ who can win the game $\mathbf{Game_2}$ with a non-negligible advantage ϵ, then there exists a simulator $\mathcal{C}_2$ that can solve the CDH problem.

Lemma 2: In the ROM, suppose that there exists a PPT adversary $\mathcal{A}_3$ who can win the game $\mathbf{Game_3}$ with a non-negligible

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advantage ϵ, then there exists a simulator $\mathcal{C}_3$ that can solve the CDH problem.

Security Analysis: This subsection offers a comprehensive examination of the diverse security features that our proposed scheme fulfills.

1) *Message Authentication:* To ensure the integrity of the message and the reliability of its source, each sender signs the message using the $\mathsf{A-Sign}$ algorithm before passing it on to the receiver. After receiving the message-signature pair, the receivers will use the $\mathsf{Ver-Dec}$ algorithm to verify the authenticity of the signature. This process can effectively prevent potential forgery and ensure the security of communication.
2) *Confidentiality:* During the communication process, there are sensitive messages that need to be kept confidential. Our scheme embeds these sensitive messages into the signature. We have provided complete proof of the indistinguishability of the ciphertext before, and the ciphertext length is consistent with σ_3. Other users in the system cannot distinguish whether σ_3 is ciphertext or part of a normal signature scheme.
3) *Non-Repudiation:* Prior to this, we have demonstrated the unforgeability of the proposed scheme. If the vehicle refuses to confirm its signature, it will be considered a malicious user, at which point the TA has the ability to track and reveal the true identity of the malicious vehicle. Therefore, the vehicle cannot refuse its signature on the message.
4) *Identity Anonymity:* $\mathcal{V}_i$ requires registration with TA before communication. During this process, TA selects a random number k and generates pseudonyms PID_i for $\mathcal{V}_i$ through the encryption algorithm. Vehicles use pseudonyms for communication in subsequent communications. Throughout the process, k is randomly selected by TA, so no one knows its corresponding true identity except for TA and itself. No one can deduce the corresponding real identity based on pseudonyms, as we use a secure encryption algorithm to ensure the anonymity of our *Sphinx*. It is worth emphasizing that our *Sphinx* is consistent with the existing CPPA, which mainly utilizes pseudonymization to achieve identity anonymity. Therefore, the possibility of obtaining real identity through additional information is not considered.
5) *Condition Tracking:* Although the vehicle uses pseudonyms PID_i for communication, it does not mean that it can do as one pleases. When the system identifies vehicles with inappropriate behavior, TA can track their true identities based on relevant pseudonyms and impose penalties on them.
6) *Unlinkability:* In communication, each vehicle has the capability to request an independent pseudonym PID_i, with each pseudonym equipped with its own private key, ensuring no direct correlation between them. Additionally, the random number used to generate the signature is selected instantaneously and at random. As a result, any potential attacker attempting to trace the true identity of a vehicle by analyzing the pseudonym or signature will be unsuccessful.

VI. Performance Evaluation

In this section, we will analyze the performance from two aspects: *theoretical complexity* and *experimental simulation*. As our proposal is being put forward for the first time, we only need to conduct experimental simulations on it.

Theoretical Complexity: Table II presents a comparison of different schemes in terms of security requirements, while Table III focuses on comparing the differences in computational and communication costs among the schemes with the closest relationship.

For security requirements, due to the unique properties of the VANETs, specific requirements often need to be met. The requirements shown in Table II are the most worth considering in VANETs, where "Addition Communication" indicates whether additional authentication messages need to be transmitted during encrypted communication. After analysis, we have divided the schemes into two categories: CPPA protocols and signcryption schemes. Among them, in the CPPA scheme, except for WCX+21, which cannot achieve the unforgeability of messages, all other schemes can achieve conditional anonymity and unforgeability. However, WCX+21, ZLL+24, XCMZ22, and MCL+23 cannot achieve message confidentiality, and although WCZ+24 can achieve message confidentiality, there is a key custody issue. XDM+23 and GGG+24 are signature encryption schemes that can achieve both unforgeability and confidentiality of messages simultaneously, but cannot achieve conditional anonymity. These two requirements are both very important for the Internet of Vehicles. And our *Sphinx* can meet all the requirements involved in VANETs, making it more suitable for practical applications.

For computation and communication costs, our solution increases the computational overhead in the signature generation process by integrating the ciphertext and the signature. However, since ciphertext becomes part of the signature, the communication cost of transmitting messages is reduced. In latency-sensitive VANETs, communication cost is one of the factors that must be focused on. To make the comparison as fair as possible, we choose the short signature scheme (such as BB signature [49]) as the "SIG" when comparing the communication overhead. As shown in Table III and Fig. 6(c), we can find that the communication cost of our *Sphinx* has been greatly reduced, making it suitable for scenarios where the capacity of a single anamorphic signature transmission channel is limited (typically 32 Bytes). Meanwhile, the increased computational overhead for this purpose is within reasonable limits. Among them, "Authentication" refers to the process of signing and encrypting, and "Verification" refers to the process of verifying the signature and decrypting. While algorithms such as signing and verifying signatures require relatively few computational resources in the MCL+23 and WCZ+24, the negotiation of the session key is involved, which greatly increases the communication cost of the schemes. In addition, although GGG+24 does not require negotiation of session keys and can achieve both signing and encrypting. However, it needs to be generated into a ciphertext first, and then the ciphertext is signed. Therefore, when transmitting messages, the signature and ciphertext need to be transmitted separately, which undoubtedly increases the communication overhead significantly.

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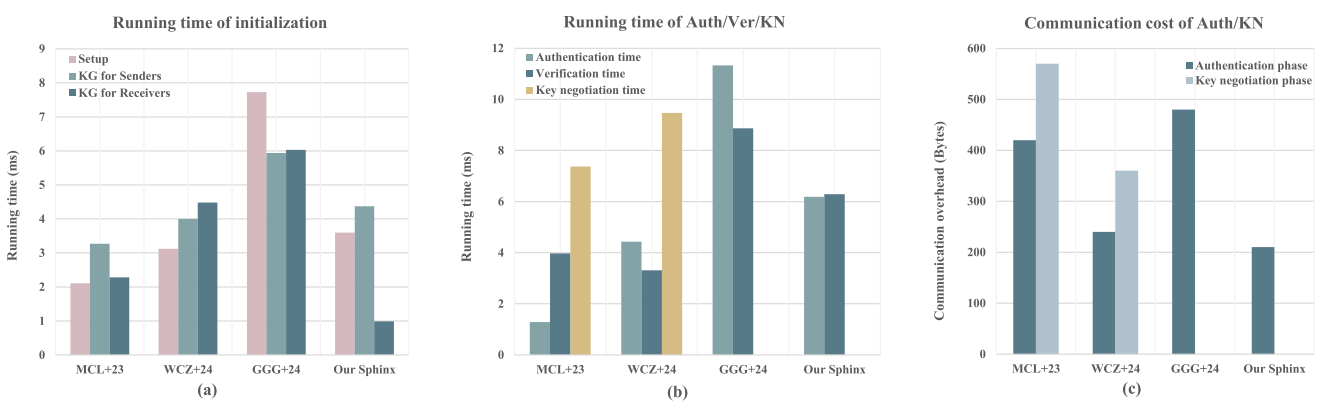
TABLE II
COMPARISONS SECURITY OF DIFFERENT AUTHENTICATION PROTOCOLS

	Identity Privacy	Traceability	Unforgeability	Confidentiality	Certificateless	Additional Communication
WCX+21 [31]	✓	✓	✗	✗	✗	✗
XCMZ22 [36]	✓	✓	✓	✗	✓	✗
MCL+23 [37]	✓	✓	✓	✗	✓	✗
XDM+23 [46]	✗	✗	✓	✓	✓	✗
WCZ+24 [42]	✓	✓	✓	✓	✗	✗
ZLL+24 [38]	✓	✓	✓	✗	✓	✗
GGG+24 [48]	✗	✗	✓	✓	✗	✗
Our Sphinx	✓	✓	✓	✓	✓	✓

TABLE III
COMPARISONS COMPUTATION AND COMMUNICATION COSTS

	Certificateless	Key Negotiation (KN)	Authentication	Verification	Total Communication Cost
MCL+23 [37]	✓	$4S_1 + 7S_2 + 4HS$	$1S_2 + 2H$ (No-Encryption)	$2S_1 + 2S_2 + 1H$ (No-Decryption)	$\|9\mathbb{G}_1 + 6\mathbb{Z}_p\|$
WCZ+24 [42]	✗	$1S_2 + 1H + 3$ (SIG+VER)	$1S_1 + 6S_2 + 4H + XOR$	$1S_1 + 3S_2 + 4H + XOR$	$\|2\mathbb{G}_1 + 2\mathbb{Z}_p + 3$ (SIG-length)$\|$
GGG+24 [48]	✗	-	$((3l + U_s + U_e + 2)N + l + 8)S_1$ $+((2N + 1)l + 9)S_2 + 5H + 1XOR$	$((2l + 1)N + 1U_e + 2)S_1$ $+(2lN + 2)S_2 + 2EP + 4H + 1XOR$	$\|5\mathbb{G}_1 + \mathbb{Z}_p\|$
Our Sphinx	✓	-	$3S_1 + 6S_2 + 2H + XOR$	$4S_1 + 4S_2 + 2H + 3EP + XOR$	$\|2\mathbb{G}_1 + \mathbb{Z}_p\|$

S_1/S_2: points addition/scalar multiplication on additive group.
U_s/U_e: signature/encryption attribute space.
SIG/VER: common signature/verification algorithms.
EP: bilinear pairing operation.
l: number of liner secret sharing.
XOR: XOR operation.
H/HS: hash/hash to string function.
N: number of attribute authorities.



Remark 3: MCL+23 cannot satisfy confidentiality. (no encryption and decryption algorithms)

Fig. 6. Experimental performance.

Experimental Simulation: For implementation, we chose the representative solutions and deployed them in a personal computer equipped with Ubuntu 18.04 virtual machine, utilizing Python 3.6 and the PBC library. The personal computer is configured with Lenovo Intel (R) Core (TM) i7-10700 CPU @ 2.90 GHz, 16.0 GB RAM, and Windows 11 22H2. Our scheme requires a symmetric linear group, therefore we heuristically use the SS512 curve, which provides a security level of 80-bit [50]. The experimental simulation results are shown in Fig. 6.

Due to some schemes involving anonymous identity generation or certificateless, there are some differences in algorithm composition, but it can roughly eliminate the overall regularity of the system and be divided into three parts: *setup*, *key generation for senders and receivers*, and *communication*.

As shown in Fig. 6(a), the running time of the programs in the *Setup* is very close to each other. This is due to the fact that the *Setup* algorithm is executed only once during initialization and does not have a significant impact on the overall system performance.

Regarding the *key generation for senders and receivers* stage, as shown in Fig. 6(a), the key generation algorithms for MCL+23, WCZ+24, and GGG+24 receivers have significantly longer running times than our *Sphinx*. Although MCL+23 and WCZ+24 are slightly faster than us in the sender's key generation, overall, our solution has a clear advantage in terms of overall runtime at this stage.

For the *communication* phase, as shown in Fig. 6(b), although our scheme is slightly inferior to MCL+23 and WCZ+24, it avoids key negotiation, which usually incurs high communication overhead. GGG+24 has no key negotiation, but it uses ABE, and the computation overhead scales linearly with the number of attributes. For a fairer comparison, we set the number

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of parameters, such as attributes, attribute authorities, etc., to a fixed value of 1. Even so, our scheme still outperforms it. It is worth noting that the authentication and verification time of MCL+23 is very short due to the fact that it cannot encrypt sensitive information. It cannot ensure the confidentiality of information, which is critical for VANETs.

Overall, considering both theoretical and experimental analysis, our solution performs better and is more suitable for practical VANETs scenarios.

VII. Conclusion

In this paper, we introduce a new cryptographic primitive, asymmetric anamorphic signature (AAS), and present the *Sphinx* scheme by integrating AAS with certificateless conditional privacy-preserving authentication (CL-CPPA). Our *Sphinx* achieves authentication, anonymity, traceability, and confidentiality without the need for additional secure transmission or complex key agreements. This is a challenge that cannot be solved by previous anamorphic signature schemes. We believe that our solution has the competitive potential to provide reliable security for vehicular ad hoc networks.

In our proposed CL-CPPA scheme, we incorporate the reconstruction anamorphic signature, intending to ensure the conditional anonymity of users while enabling efficient and secure transmission. That is, sensitive messages can be securely transmitted without the knowledge of other entities in the system, significantly enhancing the security of communication. However, with our AAS based on the Hohenberger-Waters signature construction, bilinear-pair mapping operations are required, which causes some computational overhead. In this connection, we will design a pairing-free AAS to further improve the efficiency of *Sphinx* in future work. Furthermore, the practical deployment performance of the proposed scheme will be influenced by multiple factors, including network conditions (e.g., signal interference, latency, bandwidth limitations) and vehicular scale with dynamic distribution patterns. We will conduct real-world deployment tests in future work to comprehensively validate its feasibility and robustness.

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